

1 ☐ **First: Understand What Your Instructor *Really* Wants**

Let's challenge a common mistake first ☐

Many students jump straight into Packet Tracer and “make it work”. That's **not enough**.

Your grading is based on **three things**:

1. **Correct network design**
2. **Correct configuration (OSPF + ACL)**
3. **Your understanding and documentation**

So your project has **two parallel tracks**:

- **A. Packet Tracer Implementation**
- **B. Written Documentation (steps + screenshots + explanation)**

You must build both **together**, not separately.

2 ☐ **Network Design (VERY IMPORTANT)**

Recommended Simple but Strong Topology

Do **NOT** overcomplicate. One router is enough.

```
PC1 ---- Switch1 ----|
                        |---- Router ----
Switch2 ---- PC2
PC3 ---- Switch1 ----|
```

Devices to Use

- **1 Router** (e.g. Router 2911)
- **2 Switches** (2960)
- **3–4 PCs**

Why this works:

- Multiple networks → OSPF is meaningful
- Router is central → ACL firewall makes sense
- Easy to explain during demo

3 ☐ IP Addressing Plan (Before Configuration)

Never configure without planning IPs first.

Device	Interface	IP Address	Subnet
Router	G0/0	192.168.1.1	/24
Router	G0/1	192.168.2.1	/24
PC1	NIC	192.168.1.10	/24
PC3	NIC	192.168.1.11	/24
PC2	NIC	192.168.2.10	/24

Gateway for PCs:

- Network 1 → 192.168.1.1
- Network 2 → 192.168.2.1

☐ **Screenshot this table** for your documentation.

4 ☐ Step-by-Step Packet Tracer Implementation

☐ **Step 1: Physical Setup**

1. Place router, switches, PCs
2. Use **Copper Straight-Through** cables
3. Connect:
 - PCs → Switches
 - Switches → Router

☐ Screenshot: *Physical topology*

☐ **Step 2: Configure IP Addresses on PCs**

For each PC:

- Desktop → IP Configuration
- Enter IP, subnet, gateway

☐ Screenshot: *PC IP configuration*

☐ **Step 3: Configure Router Interfaces**

On the router (CLI):

```
enable
configure terminal

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
```

```
interface g0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
exit
```

❑ Screenshot: *Router interface configuration*

5❑ Objective 1: OSPF Routing Configuration

Why OSPF?

You must be able to answer this:

“OSPF is a dynamic link-state routing protocol that automatically updates routing tables, making the network scalable and fault tolerant.”

Configure OSPF on Router

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
exit
```

Verify:

```
show ip route
```

You should see routes marked with **O**.

❑ Screenshot:

- OSPF configuration
- Routing table output

□ Devil's advocate:

If you only have one router, **why OSPF?**

Answer during demo:

“This simulates a scalable network. OSPF is implemented to demonstrate dynamic routing principles, even though the topology is small.”

That answer saves you marks.

6 □ Objective 2: Firewall Security Using ACL

Scenario (Keep It Simple)

- **Allow** PC1 to access PC2
- **Deny** PC3 from accessing PC2

Standard ACL Example

```
access-list 10 deny host 192.168.1.11
access-list 10 permit any
```

Apply ACL to router interface **towards destination network:**

```
interface g0/1
ip access-group 10 out
exit
```

Testing

- PC1 → ping PC2 ☐
- PC3 → ping PC2 ☐

☐ Screenshot:

- ACL configuration
 - Successful and failed ping tests
-

7 ☐ Documentation Structure (VERY IMPORTANT)

Your report should follow this structure:

1. Introduction

- Brief overview of the project
- Objectives (OSPF + ACL)

2. Network Topology

- Diagram (Packet Tracer screenshot)
- Devices used

3. IP Addressing Scheme

- Table (like above)

4. OSPF Configuration

- Commands used
- Explanation
- Screenshot

5. Firewall (ACL) Configuration

- Security goal
- ACL commands
- Test results

6. Conclusion

- What you learned
 - Importance of OSPF and firewall security
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8 How to Prepare for Evaluation

Each group member must answer:

- What is OSPF?
- Why area 0?
- Difference between static routing and OSPF
- Why ACL applied on **outbound**, not inbound
- What happens if ACL order is changed?